AI Project – Team M Requirements Handout

Oct 23, 2026 · @Emranul Hoque

At a glance

Team M must hand in a trained imitation-learning model that makes a Niryo Ned2 push a puck along a new, unseen black-line path. It is due by Jan 10, 2027 at 23:59, and the team presents it on Jan 11, 2027.

Team	Team M: Emranul Hoque, Marshal Sebastine (Moodle spelling)
Supervisor	Prof. Dr.-Ing. Thomas Nierhoff
Course	MAI/MKI – AI Project / KI Project (WS), winter semester 2026/27
Project	Project 6: imitation learning using a Niryo robot (a Ned2: confirmed on our unit)
Shared with	Groups G, I, N, S, AK and AZ (7 groups on Project 6, although the slide caps it at 5)

The task

The robot must learn to push a puck along a black line on the table towards a goal position, using imitation learning. Its input is the end-effector camera video and its output is robot motion. Sources: kickoff slide 11, and the announcement "Information about the AI project" (05.10.26).

1. **Demonstrate.** We push the puck ourselves using the robot, while recording the video from the webcam mounted on the end effector.
2. **Build the dataset.** The video is the model's input and the robot's movement is the output the model learns. The professor expects this, not training, to be the hard part.
3. **Train a model.** The robot's movement must come from a model trained by imitation learning on our demonstrations. In our reading, a hand-coded line follower would not meet that. For Projects 1 and 2 the professor required offline models; for Project 6 he has not said, so we ask (question 10). Training may use the GPU pool.
4. **Generalise.** The final evaluation uses a new, previously unseen path. Training on one path, or memorising a trajectory, would likely not transfer (our inference).

5. **Head for the goal.** Push the puck along the path towards the goal position. How close counts as success is question 8.

Prof. Nierhoff also gave us photos of six example path sheets on 23.10.26, from a straight line to a dashed self-crossing loop (`01_course/path_sheets`). They are not on Moodle.

Not specified yet: the action representation (joint angles or end-effector poses), the control rate, and how the evaluation is scored. These are our design choices or open questions.

Hard rules

To pass you need four things:

- the exam registered in PRIMUSS between 9 and 17 Dec 2026;
- the project work handed in and presented;
- each member's share identifiable;
- a grade of 4.0 or better.

Attending all four meetings is a separate course rule: each absence without a medical certificate costs points.

Conflict to settle on Monday: the module handbook requires a 7-page documentation with a literature review. On Moodle, the professor expected documentation only for Project 3: "Different to the other projects you'll probably have to write some documentation" (05.10.26). Which applies is question 7.

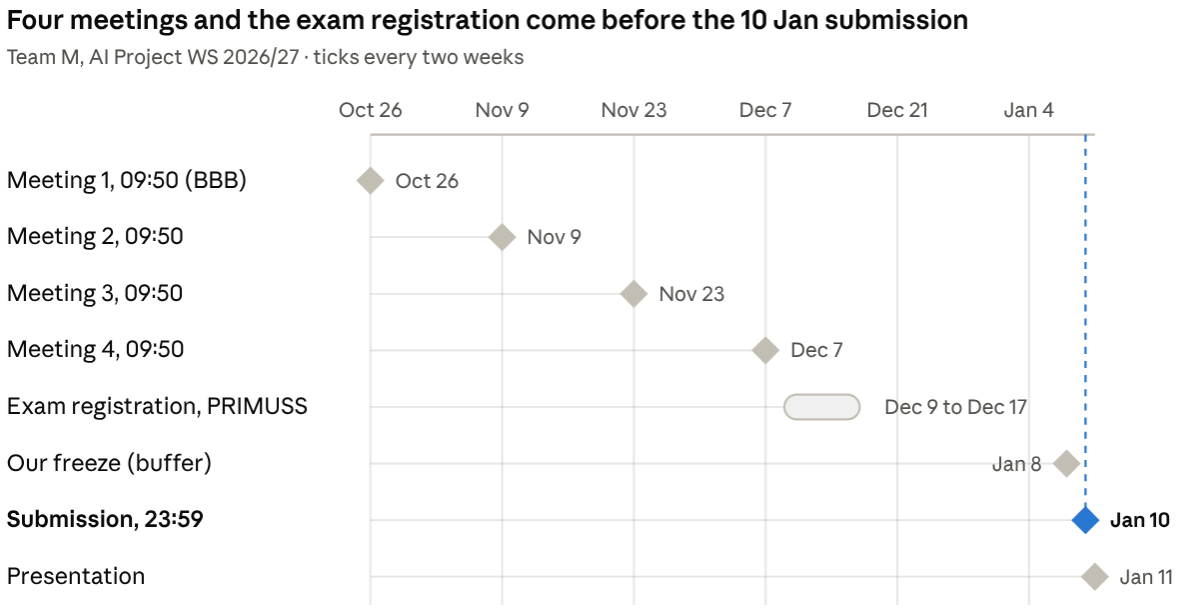
Rule	What it means for us	Source
Register for the exam	In PRIMUSS, 9–17 Dec 2026; binding. If you miss it, the exam counts as not taken: no grade this semester, and a late registration needs a justified request. If you are registered but do not submit or present, you fail, unless the reason was beyond your control and you report it to the Prüfungsamt at once (illness: a doctor's certificate from that day). Print the receipt, and check the status on the next working day.	OTH dates, OTH FAQ, ASPO §§ 18, 19, 32; announcement 05.10.26

Rule	What it means for us	Source
Attend every meeting	All four BBB meetings, 09:50–10:00. Points are deducted for each absence without a medical certificate.	Moodle timetable; announcement 22.10.26
Assessment: project work	A 15-minute team presentation plus a 7-page project documentation (see the conflict above). ASPO allows about 3–10 pages for programming tasks.	MAI Course Catalogue 2026; ASPO § 25
Theory and practice	A theoretical part (literature review) and a practical part (software programming).	MAI Course Catalogue 2026
Individual share	Each member contributes individually. Each contribution must be clearly identifiable and counts for at least 50 % of the grade. Log who did what from day one.	ASPO §§ 25, 29
Imitation learning	A trained model, learned from our own recorded demonstrations (see The task).	Kickoff slide 11, announcement 05.10.26
Grade and credit	5 ECTS (4 SWS, 150 h: 60 h contact, 90 h project work), taught in English. Graded 1.0–4.0 to pass, 5.0 to fail. A fail must be retaken within the next two semesters, and the module runs only in winter.	Course Catalogue (workload, language); MAI SPO Annex 1; ASPO §§ 34, 38
Winter semester only	Grades are submitted only in the winter semester, so finish this semester.	Kickoff slide 2

The course's official Monday slot is 08:00–11:15, room EMI301; our meeting falls inside it (MAI timetable).

Deliverables and deadlines

Everything leads to the submission on 10 Jan 2027 at 23:59. The deadline most easily missed is the exam registration, 9–17 Dec 2026.



deadlines · Oct 2026 – Jan 2027

The four meetings fall two weeks apart, and the registration window opens two days after the last one. Our own freeze on 8 Jan leaves two days of buffer before the submission. The professor must announce the task scope at least four weeks before the submission date (ASPO § 25), so the submission rules should be on Moodle by about 13 Dec 2026.

Deliverable	What it contains	Confirmed by
Exam registration	PRIMUSS entry for AI Project; print the receipt and check the status on the next working day	OTH FAQ; ASPO § 18
Meeting notes	Progress, results, problems, and next steps with an owner each; one per meeting	Our rule (`05_meetings`)
Submission package	A 7-page documentation with a literature review (theory) and a software part (practice), if the handbook applies. Code and the trained model follow from the task. Exact contents: question 7	Module handbook; kickoff slide 11. The Moodle submission activity is not published yet

Deliverable	What it contains	Confirmed by
Presentation	11 Jan 2027, "(Live)"; 15 minutes as a team. Whether the unseen-path test runs there: questions 11 and 12	Kickoff slides 3 and 5; module handbook

Robot, lab and computing rules

The Ned2 is borrowed lab equipment. These rules come from the professor's FAQ and the official Niryo manual; breaking them risks the robot and our data.

Area	Rule	Source
Placement	Use a firm, flat table larger than the 200 × 200 mm base. **Never place it near or at the table edge.** Keep the reach circle (about 500 mm radius) clear.	Prof. FAQ; Ned2 specs
Before use	Check the workspace and keep fingers away from pinch points. On long sessions, watch the motor temperature in NiryoStudio.	Ned2 manual, safety
Power on	Use the back-panel button, wait for the **blue** LED ring, then calibrate. Calibration is automatic and needed after every reboot (`calibrate_auto()`, or the Custom button).	Ned2 manual
Stopping	**PyNiryo cannot abort a motion once it is sent: each `move()` runs to completion.** For first tests, cap the speed (`set_arm_max_velocity()`), keep moves short, and keep a hand near the e-stop on the safety box. The e-stop stops all motion at once, but the arm can drop (cushioned; there are no brakes), so do not touch the robot when you press it. A long press (> 2 s) on the top button stops a NiryoStudio program.	PyNiryo API; Ned2 manual
Shutdown	Move to the home position (`move_to_home_pose()`), then press the power button. Never unplug it while it runs: this risks the SD card.	Ned2 manual
Collision	A flashing orange ring means a collision. Check for the obstacle, then start a new action. In code: `collision_detected`, `clear_collision_detected()`.	Ned2 manual; PyNiryo API

Area	Rule	Source
Moving it by hand	Learning mode (FreeMotion): hold the wrist button, or call `set_learning_mode(True)`; the ring breathes blue. Support the arm when the motors release. Built-in recording samples waypoints at only 7 Hz, so log the demonstrations ourselves.	Operating modes; ned_ros
One connection	PyNiryo allows one client at a time, and each call blocks until it is done. A separate recorder script cannot connect alongside the control script, so plan the recording around this.	ned_ros TCP server
Tools	Mount only Niryo-supplied tools; for a self-designed tool the manual says to contact Niryo support. Custom 3D-printed jaws for the Custom Gripper are explicitly allowed. Our suggestion, pending question 4: build the puck pusher as custom jaws. No hardware changes, and no edits to the robot's configured limits.	Ned2 manual
Camera	Not built in (question 2). With Niryo's Vision Set, use `get_img_compressed()` and decode with `pyniryo.vision.uncompress_image()`: 15 fps from software 5.8.0, 5 fps before. With a USB webcam on the laptop, use OpenCV.	niryo.com; PyNiryo vision
Limits	Payload 300 g; the tool tip moves at most 468 mm/s.	Ned2 specs
Puck	One per group, from the table in front of the Maker Space (DC 1.04).	Prof. FAQ
GPU pool	Room DC 1.07. Log in with your full OTH e-mail, so the address has two @ signs: `ssh first.last@oth-aw.de@ki-0xx.oth-aw.de`. The fingerprint prompt must name `ki-0xx`. Needs the OTH VPN (FortiClient) and activation by the professor; usually only the last row is on.	`01_course/gpu_pool_ssh_guide_en.pdf`; announcement 12.10.26

Still unknown

Twelve requirements are not published yet, and we ask for them meeting by meeting. Items 1–7 decide how we record data and how much we write, so they go first. The live list, with answers, is `05_meetings/requests_to_professor.md`. The professor's FAQ grows

over time, so re-check it before each meeting.

#	Question	Ask at
1	Original path-sheet files (SVG or PDF)	Meeting 1
2	Which end-effector camera, and where it is mounted	Meeting 1
3	May we print our own path sheets for training?	Meeting 1
4	Demonstrate by hand (learning mode) or by jogging? Is there a fixed pusher tool?	Meeting 1
5	Robot software version, and may we update it?	Meeting 1
6	Is our GPU-pool SSH access active?	Meeting 1
7	Does the handbook's 7-page documentation with a literature review apply, and what exactly do we submit (format, where)?	Meeting 1
8	How is the unseen-path test scored: how close counts as success?	Meeting 2
9	Test setup: sheet position, puck start, lighting	Meeting 2
10	Must the final policy run offline or locally, and are pretrained vision models allowed?	Meeting 2
11	Presentation: the handbook says 15 minutes as a team. Is there a live robot demo?	Meeting 3
12	When and where the final test runs, and when we get the test path	Meeting 4

Start now

With the robot in hand, these steps make every requirement reachable from day one. The first six are this weekend's job; the rest come before meeting 1 on Monday.

- [ ] **Install first, while you have internet:**
  - NiryoStudio;
  - a Python 3.11 environment with `pip install pyniryo==1.2.5 opencv-python`.

  The robot's hotspot has no internet.
- [ ] **Unbox and check.**

  - Note what came: the safety box with e-stop, the Custom Gripper, the Vision Set camera or none.
  - Keep the e-stop box wired between the power supply and the robot.
  - Model: Ned2 (confirmed).
- [ ] **Place it safely.** Firm table, away from the edge, with the reach circle clear.
- [ ] **Connect.**
  - Power on and wait for the blue ring.
  - Over Wi-Fi: join the hotspot whose name is printed under the base (password `niryorobot`, robot IP `10.10.10.10`). A short press on the back-panel Wi-Fi button turns the hotspot on.
  - Over Ethernet: use `169.254.200.200`. This keeps your Wi-Fi internet, but may need a change to your laptop's wired network settings.
  - Connect NiryoStudio and calibrate.
- [ ] **Check the software version** in NiryoStudio. PyNiryo 1.2.x needs 5.5.0 or newer. If it is older, use `pyniryo==1.1.2`, or ask about updating (question 5).
- [ ] **Verify from Python.** Run the official setup check, which calibrates, moves to zero and switches to learning mode. Clear the workspace first and support the arm when it releases.
- [ ] **Pick up the puck:** one per group, from the table in front of the Maker Space (DC 1.04).
- [ ] **Read the camera.** Identify it and grab frames: `get_img_compressed()` plus `uncompress_image()` for the Vision Set, or OpenCV for a USB webcam.
- [ ] **Record a test demonstration** on `path_01_straight_line`.
  - Push in learning mode while logging joint states and video yourselves.
  - Save it to `03_data/raw/<date>_session_01/` with `session_notes.md`, and note the robot's software version there.
- [ ] **Start a contribution log** (who did what, dated) for the individual 50 % share.
- [ ] **Start the literature list** for the theory part: ACT, Diffusion Policy, DAgger.
- [ ] **Add the exam registration to both your calendars** (9–17 Dec 2026). Re-check the FAQ, and bring questions 1–7 to meeting 1.

Sources

- Moodle course 3755: kickoff slides, FAQ (22.10.26), timetable, 5 announcements; copies in `01_course/`
- MAI Course Catalogue, 19.05.2026 and MAI SPO
- ASPO (4th amendment, 10.10.2024) and OTH semester dates
- Ned2 manual and PyNiryo 1.2.5